BAYAR'S
PROJECT BRIEF

Enterprise AI Sales Intelligence Operating System

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1. Key Features

- Centralize HORECA sales data into a single database.
- Generate reports for any custom date range.
- Prevent duplicate data entries.
- Maintain a complete audit trail for all data imports, including timestamps.
- Automatically recommend the most suitable visualizations for data analysis.

2. Objective

Our objective is to develop an AI-powered sales intelligence agent that enables users to interact dynamically with sales data using natural language.

The platform should allow users to:

- ➢ Query the database using simple English.
- ➢ Generate custom reports for any selected date range.
- ➢ Filter and analyze data using any variable, such as:
    - Route
    - SKU
    - Coffee Machine
    - Customer
    - Sales Representative
    - Or any other available attribute.

3. AI Capabilities

The AI agent should be capable of:

- ➢ Understanding user intent and natural language queries.
- ➢ Recognizing patterns and trends in the data.
- ➢ Adapting to different writing styles and reporting preferences.
- ➢ Generating meaningful observations and business insights.
- ➢ Accessing and utilizing metadata, including:
    - Report IDs
    - Sales information
    - Customer information
    - Alias data
    - Other relevant business metadata.

4. Data Management :

All data generated on a daily basis, along with historical records, will be stored in a centralized database to ensure consistency, accuracy, and easy access for reporting and analysis. This centralized repository enables the entire team to access day-to-day operational data, providing a reliable foundation for generating reports, identifying trends, and making informed business decisions.

The system should be capable of understanding historical trends, recognizing patterns, and performing advanced analytical calculations across any date range and any combination of variables. Users should be able to ask complex business questions in natural language, for example, *"What is the profit growth of Route 4 over the last 18 months, excluding the festival season?"* and receive accurate, data-driven insights and visualizations instantly.

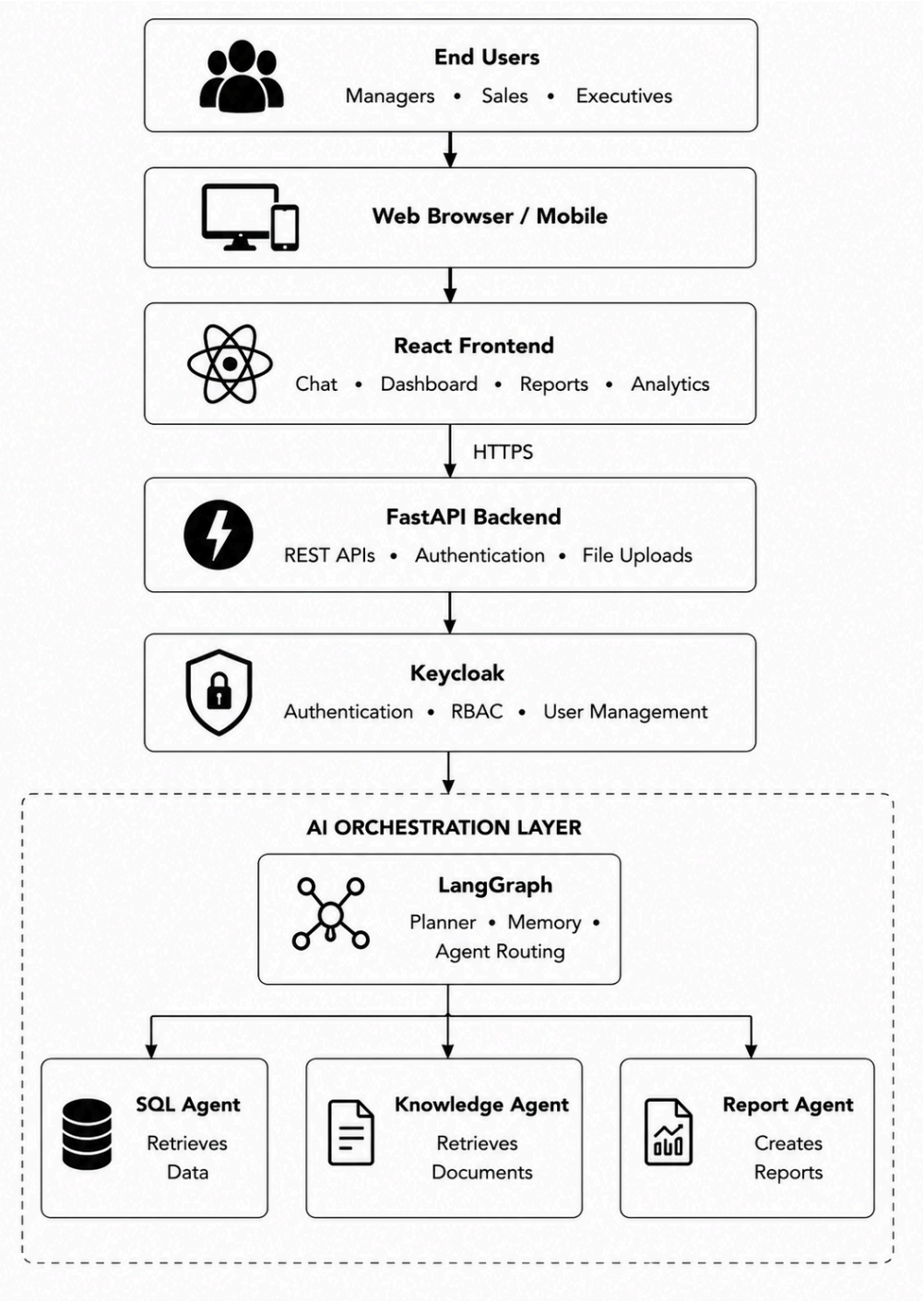
5. System Architecture :

The proposed system architecture is designed as a scalable, AI-powered business intelligence platform that combines structured business data, enterprise documents, and advanced AI models to support intelligent decision-making. The architecture is organized into multiple layers, each responsible for a specific function, including user interaction, authentication, AI orchestration, data management, analytics, and report generation.

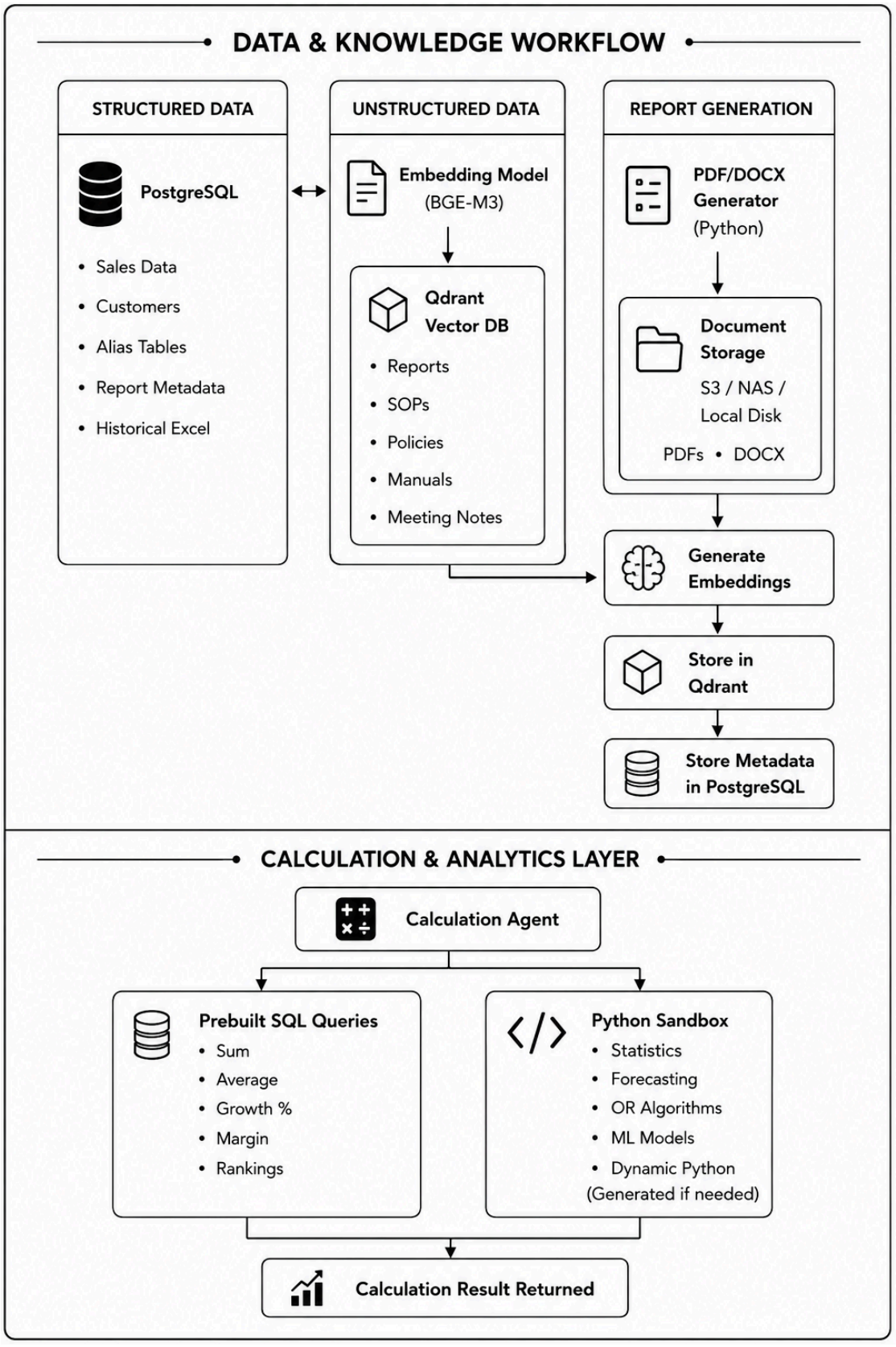
By integrating these components, the system enables users to retrieve information, perform complex business analyses, generate automated reports, and gain actionable insights through a secure and efficient workflow.

The following diagrams illustrate the overall architecture and the interaction between the different modules of the system:

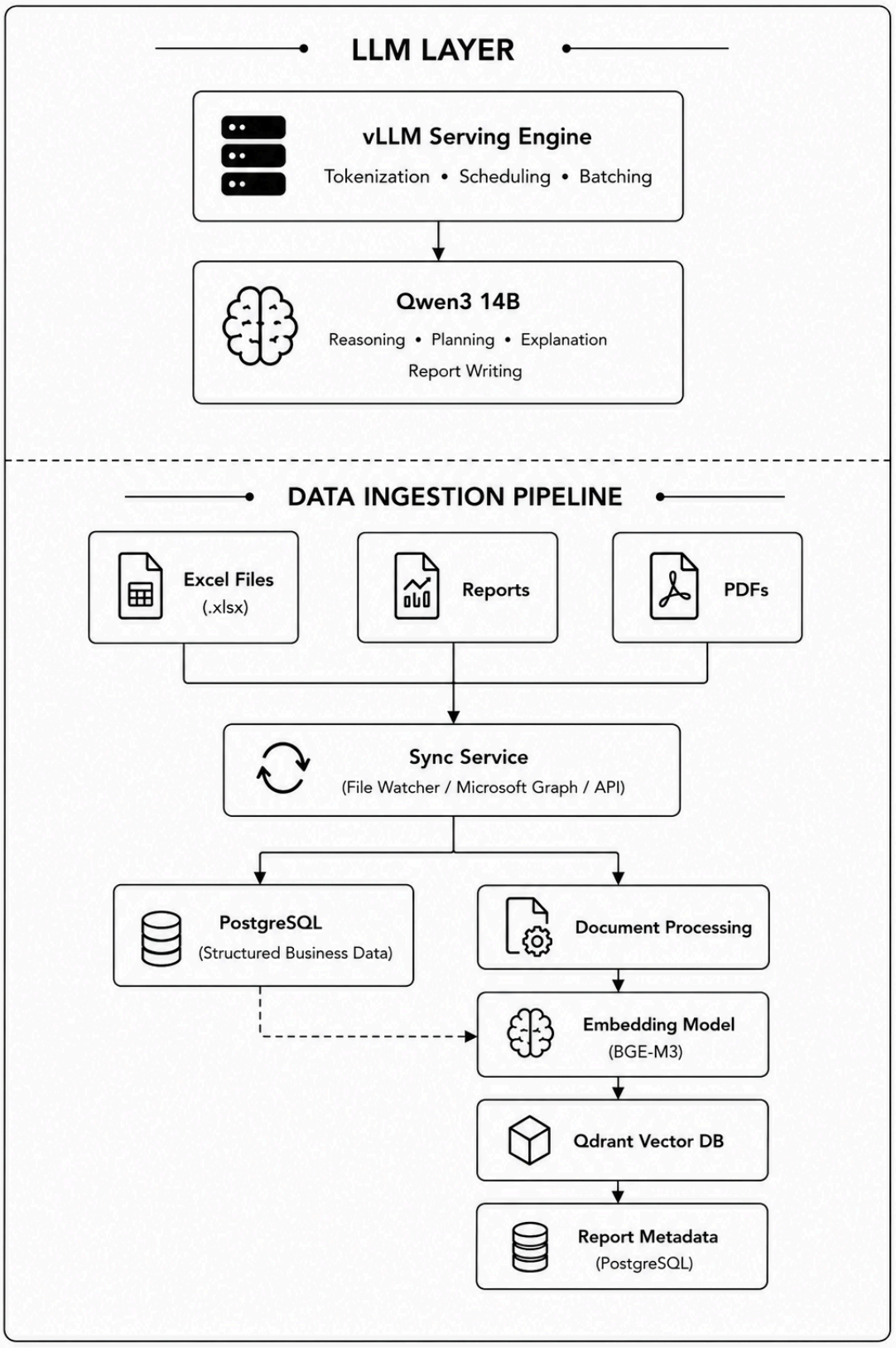
SYSTEM ARCHITECTURE



DATA AND KNOWLEDGE WORKFLOW



LLM LAYER AND DATA INGESTION PIPELINE



6. HOW A TYPICAL QUESTION FLOWS:

For example:

"What was the percentage increase in Route 4 profit compared to last month, and generate a report with recommendations."

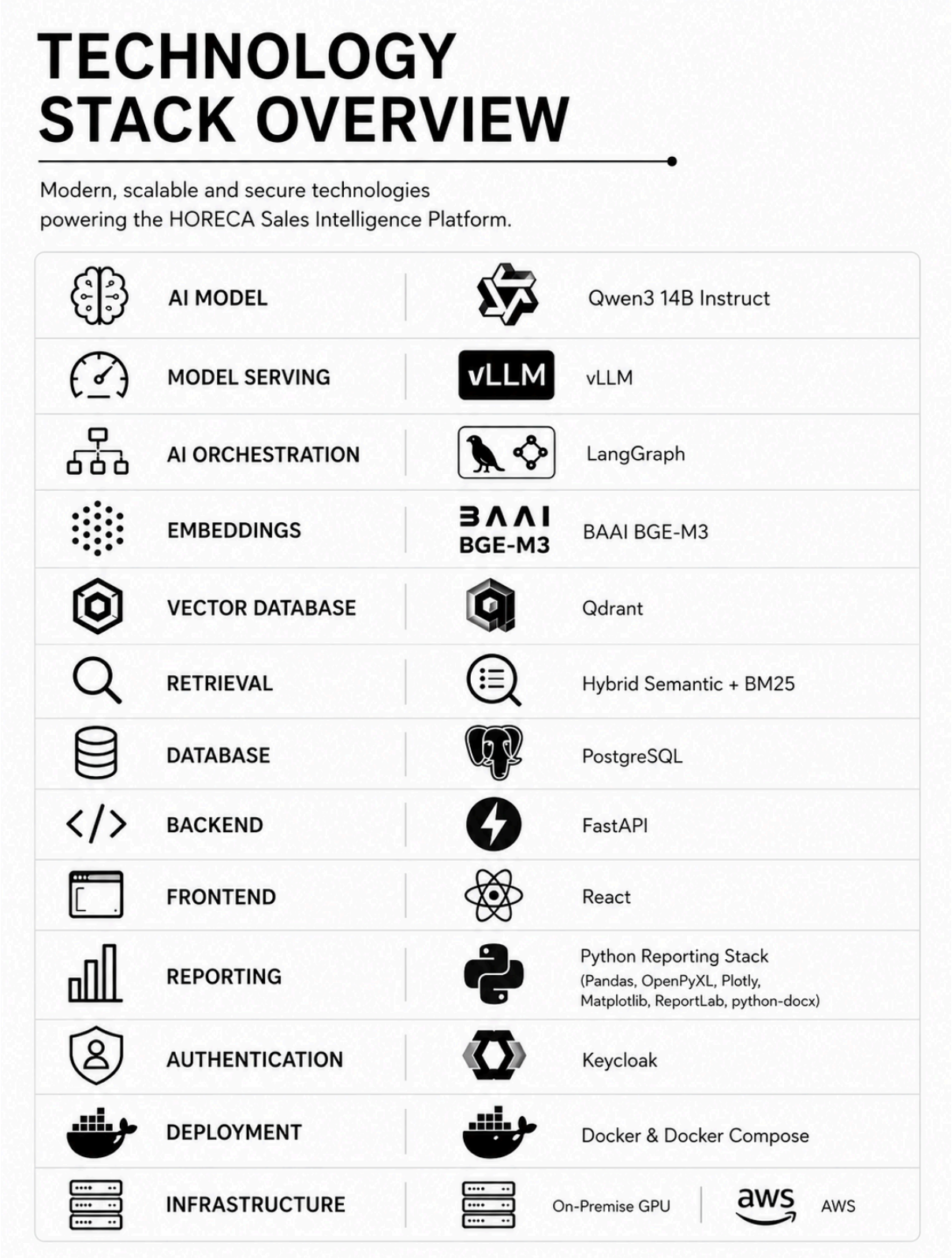
- User asks the question in the React Frontend.
- FastAPI authenticates the user via Keycloak and sends the request to LangGraph.
- LangGraph determines that it needs:
- Structured sales data → SQL Agent
- Historical reports → Knowledge Agent
- Business calculations → Calculation Agent
- The SQL Agent queries PostgreSQL.
- The Knowledge Agent retrieves relevant report sections from Qdrant.
- The Calculation Agent performs the required computations:
- Using SQL or prebuilt Python functions for common analytics.
- Using the Python Sandbox only if a novel computation requires dynamically generated code.

All retrieved facts and computed results are passed to Qwen3 14B through vLLM.
Qwen reasons over the evidence, writes the explanation, and drafts the report.
The Report Agent generates a PDF/DOCX, stores it in Document Storage, creates embeddings with BGE-M3, saves them to Qdrant, and records report metadata in PostgreSQL.
The completed response and report are returned through FastAPI to the React Frontend.

This architecture keeps each component focused on what it does best:

PostgreSQL → structured business data.
Qdrant + BGE-M3 → semantic knowledge retrieval.
Python Sandbox → safe execution of advanced calculations.
Qwen3 (served by vLLM) → reasoning, explanation, planning, and report writing.
LangGraph → orchestration and tool selection.
FastAPI + React → application interface and user experience.

7. TECHNOLOGY STACK



8. COST BREAKDOWN

➔ Option A – On-Premise GPU Server

Recommended Hardware

- NVIDIA RTX 6000 Ada or NVIDIA L40S GPU
- Sized for approximately **20 concurrent users.**

Cost Component	**Estimated cost**
Initial Hardware Investment	₹6,00,000 – ₹15,00,000
Electricity (Annual)	₹20,000 – ₹50,000
Cooling (Annual)	₹10,000 – ₹30,000
Maintenance (Annual)	₹20,000 – ₹60,000

Advantages

- Complete ownership of infrastructure
- Maximum data privacy
- Lower long-term operating costs
- Predictable performance

➔Option B – AWS Cloud Deployment

Deployment Options

- Reserved Instances
- On-Demand Instances

Cost Component	**Estimated Cost**
Monthly Infrastructure Cost	₹30,000 – ₹80,000

Estimated 5-Year Cost	₹18,00,000-,₹48,00,000

Advantages

- No upfront hardware purchase
- Easily scalable based on demand
- Pay only for resources consumed
- Managed cloud infrastructure

High-Level Cost Comparison

Deployment Model	**Estimated Cost**
On-Premise GPU Server	**₹15–20 Lakhs** (initial investment)
AWS Cloud	**₹48 Lakhs – ₹1.5 Crore** (over ~5 years, depending on usage)

Notes

- Cost estimates are based on an expected workload of **approximately 20 concurrent users**.
- AWS pricing varies depending on the selected instance type (Reserved vs. On-Demand), GPU utilization, storage requirements, and overall usage.
- The on-premise option involves a higher initial investment but typically offers lower long-term operating costs, while AWS provides greater flexibility and scalability with usage-based pricing.

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9. PROJECT MILESTONE

Phase	Time	Major Activities	Deliverable
1. Discovery & Solution Design	Week 1	Requirement gathering, process mapping, document analysis, architecture planning, infrastructure planning	Approved system architecture & implementation roadmap
2. AI Knowledge Data Base Setup	Weeks 2–3	Document ingestion pipeline, keyword indexing, embedding generation, Vector Database setup, PostgreSQL synchronization service	Searchable enterprise knowledge repository
3. AI Intelligence Layer	Weeks 4–5	LLM integration, AI Serving Engine, LangGraph workflow orchestration, Retrieval-Augmented Generation (RAG), prompt engineering	Functional AI capable of answering enterprise queries
4. Business Logic & Backend Services	Weeks 6–7	FastAPI backend, authentication (Keycloak), APIs, workflow management, session handling, report generation engine	Secure backend with enterprise APIs
5. User Interface Development	Weeks 8–9	React dashboard, chat interface, admin panel, analytics, report visualization	Interactive enterprise application
6. System Integration	Week 10	Connect frontend, backend, AI services, databases, authentication, Docker containerization	Fully integrated end-to-end system
7. Testing & Optimization	Weeks 11–12	Functional testing, AI accuracy tuning, performance optimization, debugging,	Production-ready release candidate

		security testing, user acceptance testing	
8. Production Deployment & Observation	Weeks 13–14	Deployment, monitoring, issue resolution, performance observation, feedback incorporation, documentation	Stable production deployment

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10. Project Summary

- **Estimated Duration: 12–14 Weeks**
- **Estimated Effort: 720–840 Hours**
- **Project Outcome:** Production-ready HORECA Sales Intelligence Platform

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